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Familial intracranial aneurysms

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Summary

Background We set out to determine the prevalence of incidental intracranial aneurysms in first-degree relatives aged 30 years or more of people with intracranial aneurysms, and to see if polycystic kidney disease contributes to the aggregation of familial intracranial aneurysms.

Methods 91 families with two or more affected members had previously been identified from a 14 year series of 1150 intracranial aneurysm patients treated at the University Hospital of Kuopio, Finland. Magnetic resonance angiography was used as a preliminary screening method, followed by conventional four-vessel angiography to verify suspected aneurysms. Participants were also screened for polycystic kidneys by ultrasonography.

Findings Incidental aneurysms were detected in 40 individuals: 38 of 438 individuals from 85 families without polycystic kidney disease or other diagnosed heritable disorders, and two of 22 individuals from six families known to have polycystic kidney disease. The crude and age-adjusted prevalence of incidental intracranial aneurysms among screened first-degree relatives was 8.7 (SE 1.3)% (95% CI 6.2-11.7) and 9.1 (1.4)% (6.2-11.7), respectively, for the familial group and the crude prevalence for the polycystic kidney group was 9.1 (6.1)% (1.1-29.2).

Interpretation Our results demonstrate a high prevalence of incidental intracranial aneurysms among first-degree relatives aged 30 years or older of patients with the condition and indicate that the risk of having an aneurysm is about four times higher for a close relative than for someone from the general population. Also, polycystic kidney disease families are a small fraction of the familial intracranial aneurysm families.

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Introduction

About one-quarter of cerebrovascular deaths are due to subarachnoid haemorrhage (SAH).¹ SAH is a devastating disease, since about half the patients die due to primary bleeding or to subsequent complications, and many of the survivors will need extended rehabilitation to continue an independent life. This poor outcome has changed little during the past two or three decades, even though the treatment of SAH has become a daily routine in neurosurgical centres.² The economic impact of SAH is severe because it most often affects patients in their 40s and 50s during their most productive years.³ As many as 80-90% of SAHs are caused by ruptured intracranial aneurysms (IAs).⁴⁻⁶ Up to 60% of individuals who experience aneurysmal SAH will die before hospital admission.⁶ After hospital admission, about one-third will die, about one-sixth will recover with a severe disability, about one-sixth will have some disability, and about one-third will have excellent outcome.⁶ Increasing experience in treating ruptured IAs, either with advanced microsurgical or with endovascular techniques, has resulted in improved outcomes for elective surgery of unruptured IAs.^{7,8} The per case mortality for treatment of unruptured IAs involving elective surgery is below 2%, whereas the per case mortality for treatment of ruptured aneurysms that involves emergency surgery may exceed 30% with mortality due to surgery rarely exceeding 30%.^{7,9} In addition, the techniques of endovascular surgery such as selective occlusion of aneurysms with detachable balloons and metallic coils have the promise of reducing surgical complications and mortality even further.⁸ Given the strikingly different outcomes, it is evident that treating IAs before rupture would save lives.

Screening the population at large for the presence of IAs is not feasible, but identification of groups with increased risk may make screening feasible. Familial aggregation of IAs has been observed and IAs are often associated with the autosomal dominant form of polycystic kidney disease (PKD).¹⁰ 6-20% of IA cases may be familial with two or more confirmed IA cases in the same family in the absence of any signs of other hereditary disorders.¹¹⁻¹⁴ Younger age of onset, presence of IAs among twins, and the fact that in siblings the age of onset is closer than expected by chance support the hypothesis that familial intracranial aneurysms (FIA), in the absence of other predisposing heritable disorders, are a distinct disease entity.¹⁵⁻¹⁶

The risk of at least two individuals in a family being

Location	MRA			DSA			False*	
	FIA	PKCD	Total	FIA	PKCD	Total	Pos	Neg
Internal carotid artery	5	0	5	10	0	10	0	5
Middle cerebral artery	33†	2	35	32	2	34	4	4
Anterior communicating artery	9	1	10	7	1	8	3	1
Pericallosal artery	5	1	6	7	1	8	0	2
Basilar bifurcation	4†	0	4	3	0	3	0	0
Vertebrobasilar arteries	2	0	2	3	0	3	0	1
Total	58	4	62	62	4	66	7	13

*Since individuals were screened with DSA only on the condition that they had a suspected or positive finding in MRA, it is not possible to determine the true false negatives; therefore, specificity could not be calculated and the sensitivity derived from these data is biased.

†In two individuals, one with a very small IA in the middle cerebral artery and the other with a small IA in the basilar bifurcation, IAs were not confirmed by angiography.

Table 1: Distribution of aneurysms detected by MRA and DSA according to vascular location

diagnosed with IA is unknown, primarily due to the lack of suitable methods for large-scale screening studies. Conventional angiography has been reported to be associated with an overall 4% risk of complications, about 1% risk of permanent neurological deficit, and less than 0.1% risk of mortality.¹⁷ Modern digital subtraction angiography (DSA) of the cranium is associated with a 1% transient and 0.5% permanent risk of neurological complication.¹⁸ Although the complication rates are low, the procedure is invasive and the risks are still considerable for screening symptom-free, apparently healthy individuals. Development of magnetic resonance imaging (MRI) and magnetic resonance angiography (MRA) has made possible safe, accurate, and non-invasive detection of IAs.¹⁹ Recent results on FIA families have indicated that many symptom-free family members have IAs.^{20,21}

We set out to estimate the prevalence of IAs among symptom-free relatives of FIA families, and the fraction of PCKD families among those with familial aggregation of IAs.

Patients and methods

Between 1977 and 1990, 1445 patients with cerebrovascular malformations, excluding spontaneous intracerebral haemorrhage, were treated at the University Hospital of Kuopio.⁷ This hospital serves a population of about 870 000 and, since it is the only neurosurgical unit in eastern Finland, is the site for all referrals from the catchment area that require neurosurgery. Patients with SAH who did not survive the initial bleeding events were not, however, registered in the database because they were not referred to the hospital. The University Hospital of Kuopio is also the referral centre for patients with PCKD and the departments of internal medicine and clinical genetics have a programme to track families with PCKD.

Of the 1445 patients, 1150 had IA, with (1077) or without (73) SAH.⁷ There were 92 SAH patients without IA and 82 SAH patients who were not completely studied—ie, no angiography or autopsy—among the 295 patients without IA. We obtained

Size class (mm)	FIA	PKCD	Total
Very small (≤ 6)	41	3	44
Small (7–14)	15	1	16
Large (15–24)	2	0	2
Giant (≥ 25)			
Not done*	4	0	4
Total	62	4	66

*Sizes of two aneurysms were not determined and two additional individuals with IAs estimated as very small and small by MRA were not subjected to DSA.

Table 2: Size of incidental IAs measured from DSA in the FIA and PCKD groups

information about 1067 of the 1150 patients with IA by telephone in a standardised manner asking whether there were SAH cases or inherited renal disease in the same family, as well as obtaining details of the family size up to cousins.^{11,22} Medical records of family members with suspected cerebrovascular events such as stroke or SAH were obtained and reviewed to identify the cases with known IAs. We found 91 families with at least two IA cases in each family, with a total of 207 IA patients; 85 families with 191 IA patients had no other known heritable diseases, and in six families with 16 IA patients PCKD was known to exist.¹¹

Our screening-study protocol consisted of prospective screening of symptom-free first-degree relatives (siblings, parents, and offspring) of all IA patients. We screened individuals 30 years of age and older since the mean (SD) age at onset of SAH due to ruptured IAs was 49 (13) years.¹⁵ Consequently, about 8% of aneurysmal SAH patients are under 30. Since only a fraction of the screened individuals would have an aneurysm, screening would identify very few individuals aged under 30 with IAs. After approval for an informed-consent study by the ethics committee of the University of Kuopio, potential study participants were contacted by telephone and offered the opportunity to participate. All relevant information concerning study methods, IAs, and SAH was provided at the same time. Screening took place from 1992 to 1994. We used MRA to screen for incidental IAs, and used upper-abdominal ultrasonography to screen for polycystic kidneys. Multislab three-dimensional time-of-flight MRA was done with a standard super-conducting 1.5 T whole-body system with a 10 mT/m gradient capability.²⁰ MRI of the whole brain was also done. Positive findings by MRA were confirmed by DSA if there were no contraindications.

To ensure homogeneity of the data used for comparisons, we stratified the sample into a group with PCKD and a group without PCKD or other diagnosed heritable disorders referred to as the familial form of IAs. Statistical analyses were done with Statistica for Windows (Statsoft, Tulsa, Oklahoma, USA), standard errors for Bernoulli variables were computed with a spreadsheet, and confidence intervals were computed by numerically solving for the appropriate surfaces of the tails of the binomial distribution at the appropriate significance level.

Results

FIA families

We identified 837 living, first-degree symptom-free relatives of IA patients of whom 698 were 30 years of age or older. Ultrasonography did not reveal any PCKD cases among relatives of FIA cases. MRA was not completed in 12 cases, due to claustrophobia (two), morbid obesity (three), and technical problems (seven). In addition, seven individuals had previously been screened by DSA. We completed screening on 438 of the remaining 679 individuals from 85 families for a coverage of 64.5 (SD 1.8)% (95% CI 60.8–68.1). The screened individuals consisted of 233 women, mean (SD) age 49.3 (13.8) years, and 205 men, mean age 46.9 (12.7) years. The sex distribution of the screened individuals was very similar to that found among the ruptured IA patients in the families before screening: 102 women and 89 men. The ages of the IA patients at onset of SAH were also similar: 49.3 (14.9) years ($n=102$; 11 under 30 years) for women, and 43.9 (11.8) years ($n=89$; 11 under 30 years) for men. There were no significant differences between the ages of the men or women in the screened and ruptured groups.

By MRA, 58 incidental IAs were detected in 45 individuals. Four-vessel DSA was done on 43 individuals since two women, aged 68 and 72, had advanced atherosclerosis and were considered at risk both for DSA and surgery. The IAs in these two patients were located in the right middle cerebral artery and basilar bifurcation, respectively, and were clearly visible by MRA with no

Number affected	FIA	PCKD	Total
Two	44	3	47
Three	31	1	32
Four	6	1	7
Five	1	1	2
Six	2	0	2
Seven	1*	0	1
Total	85	6	91

*The family had eight affected members after a sibling who was screened by MRA and found not to have an aneurysm developed spontaneous aneurysmal SAH.²³

Table 3: Number of affected members per family in FIA and PCKD families

potential confounding vessel loops. Consequently, these were considered true IAs. Of 43 DSAs done, seven were negative with suspected IAs being due to vessel loops, and 60 IAs were detected in the remaining 36 individuals. 13 IAs that had not been detected by MRA were detected by DSA, one of which was in the posterior inferior cerebellar artery and was not in volume studied by MRA, and another was very clearly visible upon re-examination of the MRA images. The sizes of 11 of the 13 were estimated from DSA, ten being less than 6 mm (very small) and one was between 7 and 14 mm (small). Thus, including the two aneurysms not verified by DSA, the study yielded 62 incidental IAs in 38 individuals. The location of the 62 IAs and comparison of MRA and DSA results are shown in table 1. Most of the incidental IAs were on the right side (34/62), six were in the midline, and 22 were on the left. The sizes of the aneurysms as measured from DSAs are indicated in table 2.

The study detected 36 individuals with DSA-confirmed IAs and two individuals with IAs not confirmed by DSA out of 438 screened individuals yielding a crude prevalence of incidental IAs among first-degree relatives over 30 years old of 8.7 (1.4)% (95% CI 6.2–11.7). The crude prevalence was adjusted for age with population registry data for 1990 for the catchment area, to yield an age-adjusted prevalence of 9.2 (1.4)% (6.6–12.2). One woman had aneurysmal SAH about 3 years after no IA was found by MRA.²³ Re-examination of the images did not reveal any sign of aneurysm. If she were included as affected, the crude and age-adjusted prevalences would be 8.9 (1.4)% (6.4–12.0) and 9.4 (1.4)% (6.8–12.5), respectively.

Before screening, 77.6% (66/85) of the families were known to have only two affected individuals. After screening, nearly half of the FIA families (41/85) had three or more affected family members (table 3). In FIA families the most common relationship between two affected individuals was siblings (42%); other relationships detected were that of parent-child, cousins, and aunt/uncle-niece/nephew (table 4).

Relationship	FIA	PCKD	Total
Parent-child	42	5	47
Siblings	108	6	114
Avuncular†	52	5	57
Grandparental	1	0	1
Cousins	32	6	38
Other‡	8	0	8
Total	243	22	265

*With at least two affected individuals per family, all possible relationship types between affected individuals after screening first-degree relatives are not restricted to the first degree.

†Uncle to niece/nephew or aunt to niece/nephew.

‡Relationships that do not fit the other categories—eg, half-sibs.

Table 4: Relationships between affected family members after screening*

PCKD families

Before screening six families with at least two members with IA were identified as having PCKD by history and medical records, and there were 16 individuals with IAs, of whom ten had PCKD. As noted above, no new cases of polycystic kidneys and therefore no families potentially having PCKD were found among the 85 FIA families. We identified 66 living symptom-free first-degree relatives of IA patients in the PCKD families, of whom 43 were over 30 years and 22 were screened. The study revealed two cases with aneurysms, yielding a crude prevalence of incidental IAs among the screened PCKD relatives of 9.1 (6.1)% (1.1–29.2). PCKD patients represented two of 40 (5.0 [3.5]%) [0.6–16.9]) of the newly detected IA cases. Since no new cases of PCKD were found among the FIA families, PCKD families represented six of 91 (6.6 [2.6]%) [2.5–13.8] of the families with aggregation of IAs.

MRA findings in the PCKD families are summarised in table 1, and the sizes of aneurysms in table 2. Incidental IAs were found in two individuals who were cousins: a 46-year-old woman with known PCKD had a single aneurysm, and a 46-year-old man without PCKD had three aneurysms. One patient with known PCKD had SAH while waiting for the MRA appointment, and he was operated on uneventfully. Due to the SAH, this patient was studied with DSA and it is not known if the ruptured aneurysm would have been detected by MRA. If this individual had been included in the screening the crude prevalence would have been 13.0 (3/23; 95% CI 2.8–33.6).

With respect to PCKD, an additional 11 relatives were screened with ultrasonography only. Among the 33 members of the PCKD families who had ultrasonography, 18 had normal kidneys while 15 with known PCKD had polycystic kidneys confirmed. No previously undiagnosed cases of PCKD were detected.

Discussion

The natural history of IAs is poorly understood. Long-term follow-up studies of unoperated IAs have concluded that there is a 1.4% risk of rupture per annum.^{24,25} No consistent predictor for rupture emerged from the two studies: size of aneurysm and age of patient appeared significant in one,²⁴ whereas hypertension, shape, and location were significant in the other.²⁵ Given the evident risk for a symptomless IA becoming symptomatic and rupturing, it was concluded that "an unruptured aneurysm should be operated on irrespective of size, if it is technically possible and if the concurrent diseases of the patient do not increase the surgical risk".²⁴ The operative mortality and morbidity of unruptured IAs are low, and may be lowered even further with intra-arterial surgical techniques, supporting aggressive treatment of IAs.⁸ Screening for IAs is one of the first steps towards safe and timely detection and treatment of IAs before rupture with its devastating consequences.

The development of three-dimensional computed tomography (CT) angiography and MRA has provided low-risk tools to visualise intracranial arteries and has made screening studies for IAs in symptom-free high-risk patients possible.²⁶ The disadvantages of three-dimensional CT angiography are the use of ionising radiation and contrast material. MRA is a particularly suitable method for screening studies: imaging time is short, the procedure is totally non-invasive, and it has no known risks for patient. However, compared with DSA,

MRA has less specificity and sensitivity, with the best estimates suggesting that the sensitivity to detect IAs 3 mm in size or greater approaches 95%.¹⁹⁻²¹ We compensated for decreased sensitivity by referring patients to DSA every time there was the slightest suspicion of an IA in the MRA images. Consequently, we had a high false positive rate for MRA; in seven patients no IAs were detected by DSA. Since we used MRA as an initial screening method, and patients were referred to DSA only after a suspected positive finding, we were unable to calculate a true false-negative rate. There were, however, 13 IAs, one of which was out of the study volume, not detected by MRA, the remaining 12 of which can be considered "conditional false-negative" aneurysms in individuals in whom one or more aneurysms were already detected (table 1). The occurrence of conditional false negatives may, in part, be due to the inherent lower sensitivity of MRA but also, in part, a consequence of the dual-method screening protocol—since a patient with multiple aneurysms would be referred to DSA after detection of the first aneurysm, there was a reduced need for diligence in finding all possible aneurysms after the first.

False negatives are of greatest clinical concern for any screening test. In the case of IAs and MRA, a false positive would result in unnecessary exposure to the risk of DSA whereas a false negative would result in missing a potentially lethal lesion. It is impossible to determine the number of true false negatives for MRA as used here because the study population was not screened by both MRA and DSA. Consequently, the true sensitivity of MRA on a per-aneurysm basis may be less than, but not higher than, the sensitivity based on the conditional false negatives—ie, 82% (53 IAs by both MRA and DSA, 12 conditional false negatives; $53/(53+12)$). Due to the low "sensitivity" we do not know how many IAs were missed in the MRA screening. There may be individuals who were negative in the screening but who will go on to develop aneurysms or in whom the aneurysms were formed but smaller than 3 mm in diameter and, therefore, not detectable by MRA. Indeed, one of the screened individuals in whom no aneurysms were detected by MRA developed aneurysmal SAH 3 years later.²³ Even DSA has been reported to be associated with false negatives.²⁷ The prevalence estimates here are likely underestimates of the true prevalence of IAs among the first-degree relatives of families with at least two members with IA.

Despite its shortcomings, MRA proved to be a good diagnostic and screening tool for IAs. Additionally, in the 3 years since the screening was done MRA technology has undergone rapid improvements,¹⁹ resulting in the ability to detect aneurysms smaller than 3 mm and fewer artefacts. The major disadvantages of MRA for screening studies are high cost and limited availability, which often make MRA screening of large populations unfeasible.

We screened only 438 of the 837 living first-degree relatives (53%). Because the frequency of aneurysmal SAH among individuals below 30 years of age is low, we did not think it cost-effective to screen people in this age group. The sample thus included only those 679 individuals who were 30 years or older, and we screened 438 (65%) of the available sample. The screened group matched reasonably well with the ruptured aneurysm patients from the families in terms of sex and age distribution. The low response rate of 65% could be a source of bias in the estimates if the unsampled 35% was

not randomly distributed. The response bias in terms of the inclination to participate among those apparently unaffected individuals who did or did not suspect symptoms in themselves or their apparently unaffected first-degree relatives is of greatest concern. However, this bias cannot effectively be addressed and therefore remains a potential source of bias that will result in overestimation of prevalence.

We found that the proportion of symptom-free first-degree family members who were over 30 years and had IAs was 9%, a prevalence that is at least two to four times higher than expected in a general western population.^{1,28} The ideal population prevalence for such a comparison is one derived from a screen of the same genetic and geographic population done with the same technology. Such data are not available and may be unattainable due to ethical concerns arising from the unknown and potentially harmful effects of MRA and cost considerations. Prevalence from necropsy series may have associated problems of referral bias since aneurysmal SAH cases may be over-represented, and diligent inspection of cranial scans taken post-mortem may reveal more aneurysms than are detectable in living patients. Estimates of the incidence or prevalence of SAH are more readily available and may serve as an indicator of the incidence of IAs since more than 80% of SAHs are aneurysmal. These estimates have been decreasing both for Finland²⁴ and for other western countries, probably due to changing diagnostic technology.²⁸ A recent estimate from Finland was 10.6/100 000 per year,²⁴ or only 1.4 times the mean of 7.8/100 000 per year²⁸ for other western countries. Thus, if the prevalence of IAs follows the same ratio, then the relative risk of first-degree relatives is about 2.5 to four times that of the general population.

To estimate the fraction that PCKD families represented of all FIA families in the data set, several conditions were satisfied: all families were ascertained on the condition that they had at least one member treated for IA and had at least one other member with IA; families were subsequently characterised as having PCKD or not, based on history and medical records; and all individuals that were screened by MRA were also screened by ultrasonography to ensure that there were no occult cases of PCKD in families not known to have PCKD. Ultrasound screening did not reveal any new cases of polycystic kidneys other than in the six families known to have PCKD. An active programme at University Hospital of Kuopio to identify PCKD families ensured knowledge of PCKD families in the catchment area. Altogether, PCKD families represented 7% of the families with aggregation of IAs. Among the first-degree relatives that were screened in the PCKD families 9% had IAs, a finding that agrees well with other recent screening studies.²⁹ The low number of affected and screened individuals among the PCKD families precluded additional statistical evaluation. Our study supports the finding that family members of PCKD families share the high risk of having IAs, but shows that the risk is comparable to that of the FIA families.

Recently, all the previously reported families with IA were analysed statistically including use of segregation analysis.³⁰ It was found that 79% of the families had only two affected individuals and that siblings were the most common affected kinship. Among the families presented here there was a very similar fraction with only two affected individuals before screening (78%). After

screening, however, 53% had only two affected individuals and 47% had three or more affected individuals. The segregation analysis found evidence for a major gene effect—ie, for Mendelian inheritance of IAs—however, neither dominant nor recessive models gave a substantially better fit, suggesting genetic heterogeneity.³⁰ Our findings support the hypothesis that FIA is a distinct disease that is inherited independently of other heritable disorders such as the connective tissue disorders Marfan syndrome or Ehlers Danlos syndrome type IV, or PKCK that predispose to IA. Further statistical analyses with regressive models for segregation on the complete data set should provide better estimates of the genetic and environmental components and the genetic model that fits best for IAs.

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